Regulation: R23	**II / IV - B.Tech. I - Semester**									
COMPUTER SCIENCE AND BUSINESS SYSTEMS										
COURSE STRUCTURE **(With effect from 2023-24 admitted Batch onwards)**										
Course Code	**Course Name**	**Category**	**L**	**T**	**P**	**Cr**	**C.I.E.**	**S.E.E.**	**Total Marks**	
B23BS2101	Discrete Mathematics and Graph Theory	BS	3	0	0	3	30	70	100	
B23HS2101	Universal Human Values -II : Understanding Harmony and Ethical Human Conduct	HS	2	1	0	3	30	70	100	
B23IT2101	Database Management Systems	PC	3	0	0	3	30	70	100	
B23IT2102	Object Oriented Programing through Java	PC	3	0	0	3	30	70	100	
B23IT2103	Computer Organization	PC	3	0	0	3	30	70	100	
B23IT2104	Database Management Systems Lab	PC	0	0	3	1.5	30	70	100	
B23IT2105	Object Oriented Programming through JAVA Lab	PC	0	0	3	1.5	30	70	100	
B23IT2106	Python programming	SEC	0	1	2	2	30	70	100	
B23MC2103	Fundamentals of Economics and Management	MC	2	0	0	-	30	-	30	
		TOTAL	**16**	**2**	**8**	**20**	**270**	**560**	**830**	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23BS2101	**BS**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

DISCRETE MATHEMATICS AND GRAPH THEORY

(Common to CSE, CSBS, AIML, IT, AIDS, CSG, CIC, CSIT)

Course Objectives: Students are expected to

1.	Understand the mathematical arguments using logical connectives and quantifiers and verify the validity of arguments using propositional, predicate logic and truth tables.
2.	Understand various types of relations and discuss various properties of the relations
3.	Know about the concepts of counting techniques and how to solve the recurrence relations.
4.	Understand the concepts in graphs and trees.

Course Outcomes: At the end of the course, Students will be able to

S.No	Outcome	Knowledge Level
1.	**Use** the concepts of propositional and predicate logic to verify the arguments for their validity.	K3
2.	**Apply** the knowledge of set theory to understand relations, functions and their properties.	K3
3.	**Solve** different counting problems and recurrence relations.	K3
4.	**Use** the concepts of graphs and their representations.	K3
5.	**Determine** different multi graphs and tree structures.	K3

SYLLABUS

UNIT-I (10Hrs)	**Mathematical Logic:** Propositional Calculus: Statements and Notations, Connectives, Well Formed Formulas, Truth Tables, Tautologies, Equivalence of Formulas, Duality Law, Tautological Implications, Normal Forms, Theory of Inference for Statement Calculus, Consistency of Premises, Indirect Method of Proof, Predicate Calculus: Predicates, Predicative Logic, Statement Functions, Variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.
UNIT-II (10 Hrs)	**Set Theory:** **Sets**: Operations on Sets, Principle of Inclusion-Exclusion, **Relations**: Properties, Operations, Partition and Covering, Transitive Closure, Equivalence, Compatibility and Partial Ordering, Hasse Diagrams, Lattice and its Properties. **Functions**: Bijective, Composition, Inverse, Permutation, and Recursive Functions.
UNIT-III (12Hrs)	**Combinatorics and Recurrence Relations:** Basis of Counting, Permutations, Permutations with repetitions, Circular and Restricted Permutations, Combinations, Restricted Combinations, Binomial and Multinomial Coefficients and Theorems. **Recurrence Relations:** Generating Functions, Function of Sequences, Partial Fractions,

	Calculating Coefficient of Generating Functions, Recurrence Relations, Formulation as Recurrence Relations, Solving Recurrence Relations by Substitution and Generating Functions, Method of Characteristic Roots, Solving Inhomogeneous Recurrence Relations
UNIT-IV (10 Hrs)	**Graph Theory:** Basic Concepts, Graph Theory and its Applications, Subgraphs, Graph Representations: Adjacency and Incidence Matrices, Isomorphic Graphs, Paths and Circuits, Eulerian and Hamiltonian Graphs.
UNIT-V (08Hrs)	**Multi Graphs:** Multi graphs, Bipartite and Planar Graphs, Euler's Theorem, Graph Coloring and Covering, Chromatic Number, Trees and their properties, Spanning Trees-BFS and DFS Spanning Trees, Prim's and Kruskal's Algorithms.

Textbooks:	
1.	Discrete Mathematical Structures with Applications to Computer Science, J. P. Tremblay and P. Manohar, Tata McGraw Hill.
2.	Discrete Mathematics for Computer Scientists and Mathematicians, J. L. Mott, A. Kandel and T. P. Baker, 2nd Edition, Prentice Hall of India.
Reference Books:	
1.	Elements of Discrete Mathematics-A Computer Oriented Approach, C. L.Liu and D. P. Mohapatra, 3rd Edition, Tata McGraw Hill.
2.	Theory and Problems of Discrete Mathematics, Schaum's Outline Series, Seymour Lipschutz and Marc Lars Lipson, 3rd Edition, McGraw Hill.
3.	Discrete Mathematical Structures, Bernand Kolman, Robert C. Busby and Sharon Cutler Ross, PHI.
4.	Discrete Mathematics, S. K. Chakraborthy and B.K. Sarkar, Oxford, 2011.
5.	Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K. H. Rosen, 7th Edition, Tata McGraw Hill.
e-Resources :	
1.	https://nptel.ac.in/courses/106105192
2.	https://archive.nptel.ac.in/courses/111/106/111106102/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS2101	**HS**	**2**	**1**	**--**	**3**	**30**	**70**	**3 Hrs.**

UNIVERSAL HUMAN VALUES-II: UNDERSTANDING HARMONY AND ETHICAL HUMAN CONDUCT

(Common to all Programmes of Engineering)

Course Objectives: The objective of this course is to make the student aware of

1	Essential complementarity between 'Values' and 'Skills' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
2	Harmony in the human being, family, society and nature/existence
3	Holistic perspective towards life, profession and happiness

Course Outcomes: At the end of this course student will be able to

S. No.	Outcome	Knowledge Level
1	**Explain** the role of value education in achieving basic human aspirations.	K2
2	**Summarize** needs to obtain harmony in self(I).	K2
3	**Describe** criteria for human-human relationship and harmony in society	K2
4	**Explain** four orders of nature and our existence	K2
5	**Interpret** significance of harmony in holistic development	K2

SYLLABUS

UNIT-I (9 Hrs)	**Introduction to Value Education:** Understanding Value Education- Need, Basic Guidelines, Content and Process for Value Education Purpose and motivation for the course. Self-exploration as the Process for Value Education - Sharing about Oneself. Myers-Briggs Type Indicator (MBTI) Personality Test. Continuous Happiness and Prosperity – the Basic Human Aspirations and their Fulfilment. Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) - Exploring Human Consciousness. Happiness and Prosperity – Current Scenario. Method to Fulfil the Basic Human Aspirations - Exploring Natural Acceptance- understanding and living in harmony at various levels.
UNIT-II (9 Hrs)	**Harmony in the Human Being:** Understanding Human being as the Co-existence of the Self and the Body. Distinguishing between the Needs of the Self and the Body - Exploring the difference of Needs of Self (I) and Body (Happiness and Physical Facility). The Body as an Instrument of the Self (I)' (I being the doer, seer and enjoyer).

	Understanding Harmony in the Self(I) - Exploring Sources of Imagination in the Self(I). Harmony of the Self (I) with the Body (characteristics and activities of 'I' and harmony in 'I'). Programme to ensure self-regulation(*Sanyam*) and Health(*Swasth*)- Exploring Harmony of Self (I) with the Body.
UNIT-III **(9 Hrs)**	**Harmony in the Family and Society:** Harmony in the Family – the Basic Unit of Human Interaction. 'Trust' – the Foundational Value in Relationship- Exploring the Feeling of Trust- intention and competence. 'Respect' – as the Right Evaluation- Exploring the Feeling of Respect. Other Feelings, Justice in Human-to-Human Relationship. Understanding Harmony in the Society- (society being an extension of family). Vision for the Universal Human Order- Exploring Systems to fulfil Human Goal.
UNIT-IV **(6 Hrs)**	**Harmony in the Nature/Existence:** Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four orders of nature- recyclability and self-regulation in nature. Realizing Existence as Co-existence at All Levels - Understanding Existence as Co-existence of mutually interacting units in all pervasive space. Holistic perception of harmony at all levels of existence.
UNIT-V **(9 Hrs)**	**Implications of the Holistic Understanding – a Look at Professional Ethics:** Natural acceptance of human values. Definitiveness of (ethical) human conduct. A Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics: a. Ability to utilize the professional competence for augmenting universal human order b. Ability to identify the scope and characteristics of people friendly and eco-friendly production systems, c. Ability to identify and develop appropriate technologies and management patterns for above production systems. Holistic technologies, production systems and management models- typical case studies. Strategies for transition towards value based life and profession (from the present state to Universal Human Order): a. At the level of individual: as socially and ecologically responsible engineers, technologists and managers, b. At the level of society: as mutually enriching institutions and organizations.

Text Books

1.	R R Gaur, R Sangal, G P Bagaria. "Human Values and Professional Ethics", Excel Books, New Delhi, 2010
2.	R R Gaur, R Asthana, G P Bagaria. "Teachers' Manual for A Foundation Course in Human Values and Professional Ethics", 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-53-2

Reference Books:	
1.	Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.
2.	Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3.	The Story of Stuff (Book).
4.	The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi
5.	Small is Beautiful - E. F Schumacher
6.	Slow is Beautiful - Cecile Andrews
7.	Economy of Permanence - J C Kumarappa
8.	Bharat Mein Angreji Raj – Pandit Sunderlal
9.	Rediscovering India - by Dharampal
10.	Hind Swaraj or Indian Home Rule - by Mohandas K. Gandhi
11.	India Wins Freedom - Maulana Abdul Kalam Azad
12.	Vivekananda - Romain Rolland (English)
e-Resources	
1.	https://fdp-si.aicte-india.org/UHV%20II%20Teaching%20Material/UHV%20II%20Lecture%2023-25%20Ethics%20v1.pdf
2.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Note.php
3.	https://fdp-si.aicte-india.org/download/FDPTeachingMaterial/3-days%20FDP-SI%20UHV%20Teaching%20Material/Day%203%20Handouts/UHV%203D%20D3-S2A%20Und%20Nature-Existence.pdf

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2101	PC	3	0	0	3	30	70	3 Hrs.

DATABASE MANAGEMENT SYSTEMS

(Common to IT, AIDS and CSBS)

Course Objectives: The main objective of the course is to

1.	Introduce database management systems and to give a good formal foundation on the relational model of data and usage of Relational Algebra.
2.	Introduce the concepts of SQL as a universal Database language.
3.	Demonstrate the principles behind systematic database design approaches by covering conceptual design, logical design through normalization
4.	Provide an overview of physical design of a database system, by discussing Database indexing techniques and storage techniques

Course Outcomes: After completion of the course, the student will be able to

S. No	Outcome	Knowledge Level
1.	**Describe** database fundamental concepts.	K2
2.	**Apply** E-R and Relational models for creating databases.	K3
3.	**Apply** SQL features to create, manipulate and retrieve databases.	K3
4.	**Apply** normalization concepts to refine relational databases.	K3
5.	**Illustrate** transaction management concepts.	K2
6.	**Apply** indexing concepts for searching databases.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Introduction:** Database system, Characteristics (Database Vs File System), Database Users, Advantages of Database systems, Database applications. Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Three tier schema architecture for data independence; Database system structure, environment, Centralized and Client Server architecture for the database. **Entity Relationship Model:** Introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, sub classes, super class, inheritance, specialization, generalization using ER Diagrams.
UNIT-II (08 Hrs)	**Relational Model:** Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance, Relational Algebra, Relational Calculus. **BASIC SQL:** Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update).

UNIT-III (10 Hrs)	**Structured Query Language:** Basic SQL Querying Using Select and Where clauses, Arithmetic & Logical Operations, SQL Functions (Date and Time, Numeric to String Conversion). Creating Tables with Relationship, Implementation of Key and Other Integrity Constraints, Set Operations, Nested Queries, Sub Queries, Grouping, Aggregation, Ordering, Implementation of Various Types of Joins, Views (Updatable and Non-Updatable), relational set operations. [CO3, K3]
UNIT-IV (10 Hrs)	**Schema Refinement (Normalization):** Purpose of Normalization or Schema Refinement, Concept of Functional Dependency, Normal Forms Based on Functional Dependency, Lossless Join and Dependency Preserving Decomposition, (1NF, 2NF and 3 NF), Concept of Surrogate Key, Boyce-Codd Normal Form (BCNF), Multi Valued Dependencies, Fourth Normal Form(4NF), Fifth Normal Form (5NF). [CO4, K3]
UNIT-V (12 Hrs.)	**Transaction Management:** Transaction State, ACID properties, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for Serializability, lock based, time stamp based, optimistic, concurrency protocols, Deadlocks, Failure Classification, Storage, Recovery and Atomicity, Recovery algorithm. [CO5, K2] Introduction to Indexing Techniques: B+ Trees, Operations on B+ Trees, Hash Based Indexing: [CO6, K3]

Text Books:	
1.	Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke, TMH.
2.	Database System Concepts,5th edition, Silberschatz, Korth, Sudarsan,TMH.
Reference Books:	
1.	Introduction to Database Systems, 8thedition, C J Date, Pearson.
2.	Database Management System, 6th edition, Ramez Elmasri, Shamkant B. Navathe, Pearson
3.	Database Principles Fundamentals of Design Implementation and Management, Corlos Coronel, Steven Morris, Peter Robb, Cengage Learning.
e-Resources	
1.	https://nptel.ac.in/courses/106/105/106105175/
2.	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01275806667282022456_shared/overview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2102	PC	3	--	--	3	30	70	3 Hrs.

OBJECT ORIENTED PROGRAMMING THROUGH JAVA

(Common to IT, AIDS and CSBS)

Course Objectives: Students are expected to

1.	To identify Java language components and how they work together in applications.
2.	To learn the fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries.
3.	To learn how to extend Java classes with inheritance and dynamic binding and how to use exception handling in Java applications.
4.	To understand how to design applications with threads in Java.
5	To understand how to use Java APIs for program development.

Course Outcomes: By the end of the course, the student will be able to:

S.No	Outcome	Knowledge Level
1.	**Develop** simple Java programs that incorporate fundamental programming elements.	K3
2.	**Apply** the concepts of Object-Oriented Programming such as classes and objects to design java programs	K3
3.	**Apply** the concepts of inheritance and interfaces to achieve multiple inheritance	K3
4.	**Apply** the concepts of multithreading and exceptions to design multithreaded error free programs	K3
5.	**Model** event driven GUI applications which connect with databases.	K4

SYLLABUS

UNIT-I (10Hrs)	**Program Structure in Java:** Introduction, Writing Simple Java Programs, Elements or Tokens in Java Programs, Java Statements, Command Line Arguments, User Input to Programs, Escape Sequences Comments, Programming Style. **Input/Output operations:** Reading input from console (Scanner). **Data Types, Variables, and Operators :**Introduction, Data Types in Java, Declaration of Variables, Data Types, Type Casting, Scope of Variable Identifier, Literal Constants, Symbolic Constants, Formatted Output with printf() Method, Static Variables and Methods, Attribute Final, Introduction to Operators, Precedence and Associativity of Operators, Assignment Operator, Basic Arithmetic Operators, Increment and Decrement Operators, Ternary Operator, Relational Operators, Boolean Logical Operators, Bitwise Logical Operators. **Control Statements:** Introduction, if Expression, Nested if Expressions, if–else Expressions, Ternary Operator, Switch Statement, Iteration Statements, while Expression,

	do– while Loop, for Loop, Nested for Loop, For–Each for Loop, Break Statement, Continue Statement.
UNIT-II (10 Hrs)	**Classes and Objects:** Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, Assigning One Object to Another, Access Control for Class Members, Accessing Private Members of Class, Constructor Methods for Class, Overloaded Constructor Methods, Final Class and Methods, Passing Arguments by Value and by Reference, Usage of keyword this. **Wrapper classes:** Auto boxing and unboxing. **Methods:** Introduction, Defining Methods, Overloaded Methods, Overloaded Constructor Methods, Class Objects as Parameters in Methods, Access Control, Recursive Methods, Nesting of Methods, Attributes Final and Static.
UNIT-III (10 Hrs)	**Arrays**: Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays, Operations on Array Elements, Assigning Array to Another Array, Dynamic Change of Array Size, Sorting of Arrays, Search for Values in Arrays, Class Arrays, Two-dimensional Arrays, Arrays of Varying Lengths, Three dimensional Arrays, Arrays as Vectors and lists. **Inheritance:** Introduction, Process of Inheritance, Types of Inheritances, Universal Super Class-Object Class, Inhibiting Inheritance of Class Using Final, Access Control and Inheritance, Multilevel Inheritance, Application of Keyword Super, Constructor Method and Inheritance, Method Overriding, Dynamic Method Dispatch, Abstract Classes. **Interfaces:** Introduction, Declaration of Interface, Implementation of Interface, Multiple Interfaces, Nested Interfaces, Inheritance of Interfaces, Default Methods in Interfaces, Static Methods in Interface, Functional Interfaces, Annotations.
UNIT-IV (10 Hrs)	**Packages:** Introduction, Defining Package, Importing Packages and Classes into Programs, Access Control, adding a public class and non-public class to an existing user defined package, adding an interface to a user defined package, Different ways of importing packages. **Exception Handling:** Introduction, Keywords throws and throw, try, catch, and finally Blocks, Multiple Catch Clauses, Class Throwable, Custom Exceptions, Nested try and catch Blocks. **Multithreaded Programming:** Introduction, Thread Class, Main Thread- Creation of New Threads, Thread States, Runnable Interface, Thread Priority-Synchronization. Java I/O and File: Java I/O API, standard I/O streams, types, Byte streams, Character streams, Scanner class, Files in Java
UNIT-V (10 Hrs)	**String Handling in Java:** Introduction, Interface Char Sequence, Class String, Methods for Extracting Characters from Strings, Comparison, Modifying, Searching; Class String Buffer. **Java Database Connectivity:** Introduction, JDBC Architecture, Installing MySQL and MySQL Connector/J, JDBC Environment Setup, Establishing JDBC

	Database Connections, ResultSet Interface. **Java FX GUI:** Java FX Scene Builder, Java FX App Window Structure, displaying text and image, event handling, laying out nodes in scene graph, mouse events.
Textbooks:	
1.	JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
2.	Joy with JAVA, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
3.	JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4^{th} Edition, Pearson
Reference Books:	
1.	The complete Reference Java, 11thedition, Herbert Schildt,TMH
2.	Murach's Java Programming, Joel Murach
e-Resources	
1.	https://nptel.ac.in/courses/106/105/106105191/
2.	https://www.geeksforgeeks.org/java/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2103	**ES**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

COMPUTER ORGANIZATION

(Common to IT and CSBS)

Course Objectives: Students are expected to learn

1.	Principles and the Implementation of Computer Arithmetic
2.	Operation of CPUs including RTL, ALU, Instruction Cycle, and Busses
3.	Functionality of central processing unit and control units
4.	Memory System and I/O Organization

Course Outcomes: After completion of the course, the student will be able to

S. No	Outcome	Knowledge Level
1	**Identify** set of digital components, functional components and micro-operations in a basic computer system.	K3
2	**Demonstrate** various instructions and arithmetic operations	K3
3	**Illustrate** knowledge of functional components on central processing unit and various control units.	K2
4	**Determine** different memory components in a computer for better memory organization	K3
5	**Explain** different ways of communication with I/O devices and standard I/O interface	K2

SYLLABUS

UNIT-I (10 Hrs)	**Introduction:** Basic Logic functions, Logic gates, Boolean functions, Canonical forms, Simplification of Boolean functions (up to 4 variable), Basics of Flipflops, Registers, DeCourse Coders and multiplexers. **Basic Structure of Computers**: Computer Types, Functional units, Basic operational concepts, Bus structures. **Register Transfer and Micro operations**: Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro operations, Logic Micro operations, Shift Micro operations, Arithmetic Logic Shift Unit
UNIT-II (08 Hrs)	**Basic Computer Organization and Design**: Instruction Course Codes, Computer Register, Computer Instructions, Instruction Cycle, Memory – Reference Instructions. Input –Output and Interrupt, Complete Computer Description **Computer Arithmetic**: Addition and Subtraction of Signed Numbers, Design of Fast Adders, Multiplication of Positive Numbers, Signed-operand Multiplication, Fast Multiplication, Integer Division, Floating-Point Numbers and Operations

UNIT-III (10 Hrs)	**Central Processing Unit**: General Register Organization, STACK Organization. Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Execution of a Complete Instruction, Multiple-Bus Organization, **Micro programmed Control**: Control Memory, Address Sequencing, Micro Program example, Hardwired Control and Micro programmed Control.
UNIT-IV (10 Hrs)	**The Memory Organization:** Memory Hierarchy, Main memory, Auxiliary memory, Associate Memory, Cache Memory, and Virtual memory, Memory Management Requirements, Secondary Storage.
UNIT-V (12 Hrs.)	**Input / Output Organization:** Accessing I/O Devices, Interrupts, Processor Examples, modes of transfers, Direct Memory Access, Buses, Interface Circuits, Standard I/O Interfaces.

Text Books:	
1.	Computer System Architecture M. M. Mano:, 3rd ed., Prentice Hall of India, New Delhi, 1993
2.	Digital Design, 6th Edition, M. Morris Mano, Pearson Education.
3.	Computer Organization, Carl Hamacher, ZvonkoVranesic, SafwatZaky, 5/e, McGraw Hill,2002.
Reference Books:	
1.	Computer Organization and Architecture, William Stallings, 6/e, Pearson,2006.
2.	Structured Computer Organization, Andrew S. Tanenbaum, 4/e, Pearson,2005.
3.	Fundamentals of Computer Organization and Design, Sivarama P. Dandamudi, Springer, 2006.
e-Resources	
1.	https://nptel.ac.in/courses/106/105/106105163/
2.	http://www.cuc.ucc.ie/CS1101/David%20Tarnoff.pdf

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2104	**PC**	**--**	**--**	**3**	**1.5**	**30**	**70**	**3 Hrs.**

DATABASE MANAGEMENT SYSTEMS LAB

(Common to IT and CSBS)

Course Objectives: The major objective of this course is to

1	Populate and query a database using SQL DDL/DML Commands.
2	Declare and enforce integrity constraints on a database.
3	Writing Queries using advanced concepts of SQL.
4	Programming PL/SQL including procedures, functions, cursors and triggers.

Course Outcomes: After completing the course, student will be able to:

S.No	Outcome	Knowledge Level
1	**Apply** SQL to create and manipulate relational databases using Oracle 11g.	K3
2	**Apply** SQL to write complex queries.	K3
3	**Apply** PL/SQL concepts to write triggers, procedures, and programs.	K3
4	**Apply** SQL concepts to establish database connectivity through high level programs.	K3

SYLLABUS

1	Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables), examples using SELECT command.
2	Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSET, Constraints. Example:- Select the roll number and name of the student who secured fourth rank in the class.
3	Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.
4	Queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)
5	i) Create a simple PL/SQL program which includes declaration section, executable section and exception handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found) ii) Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.
6	Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.

7	Program development using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT–IN Exceptions, User defined Exceptions, RAISE_APPLICATION_ERROR.
8	Programs development using creation of procedures, passing parameters IN and OUT of PROCEDURES.
9	Program development using creation of stored functions, invoke functions in SQL Statements and write complex functions.
10	Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables.
11	Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers.
12	Create a table and perform the search operation on table using indexing and non-indexing techniques.
13	Write a Java program that connects to a database using JDBC.
14	Write a Java program to connect to a database using JDBC and insert values into it.
15	Write a Java program to connect to a database using JDBC and delete values from it.
Reference Books:	
1	Oracle: The Complete Reference by Oracle Press.
2	Nilesh Shah, "Database Systems Using Oracle", PHI, 2007
3	Rick F Vander Lans, "Introduction to SQL", Fourth Edition, Pearson Education, 2007

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2105	**PC**	**--**	**--**	**3**	**1.5**	**30**	**70**	**3 Hrs.**

OBJECT ORIENTED PROGRAMMING THROUGH JAVA LAB

(Common to IT, AIDS and CSBS)

Course Objectives: The major objective of this course is to

1	Practice object-oriented programming in the Java programming language.
2	Implement Classes, Objects, Methods, Inheritance, Exception, Runtime Polymorphism, User defined Exception handling mechanism.
3	Illustrate inheritance, Exception handling mechanism, JDBC connectivity.
4	Construct Threads, Event Handling, implement packages, Java FX GUI.

Course Outcomes: By the end of the course, the student will be able to

S.No	Outcome	Knowledge Level
1	**Apply** control statements, operators, strings and arrays to design java programs	K3
2	**Apply** Object Oriented Programming concepts to design java programs	K3
3	**Design** multithreaded and error free java programs by applying the concepts of Exception Handling and Multithreading	K4
4	**Design** interactive GUI programs by applying the concepts of JAVAFX and JDBC	K4

SYLLABUS

1	a) Write a JAVA program to display default value of all primitive data type of JAVA b) Write a java program that display the roots of a quadratic equation $ax^2+bx=0$. Calculate the discriminate D and basing on value of D, describe the nature of root.
2	a) Write a JAVA program to search for an element in a given list of elements using binary search mechanism. b) Write a JAVA program to sort for an element in a given list of elements using bubble sort c) Write a JAVA program using StringBuffer to delete, remove character.
3	a) Write a JAVA program to implement class mechanism. Create a class, methods and invoke them inside main method. b) Write a JAVA program implements method overloading. c) Write a JAVA program to implement constructor. d) Write a JAVA program to implement constructor overloading.
4	a) Write a JAVA program to implement Single Inheritance b) Write a JAVA program to implement multi-level Inheritance c) Write a JAVA program for abstract class to find areas of different shapes
5	a) Write a JAVA program give example for "super" keyword. b) Write a JAVA program to implement Interface. What kind of Inheritance can be achieved? c) Write a JAVA program that implements Runtime polymorphism

6	a) Write a JAVA program that describes exception handling mechanism b) Write a JAVA program Illustrating Multiple catch clauses c)Write a JAVA program for creation of Java Built-in Exceptions d)Write a JAVA program for creation of User Defined Exception
7	a) Write a JAVA program that creates threads by extending Thread class. First thread display “Good Morning “every 1 sec, the second thread displays “Hello “every 2 seconds and the third display “Welcome” every 3 seconds, (Repeat the same by implementing Runnable) b) Write a program illustrating **is Alive** and **join ()** c) Write a Program illustrating Daemon Threads.
8	**a)** Write a JAVA program that import and use the user defined packages **b)** Without writing any Course Code, build a GUI that display text in label and image in an ImageView (use JavaFX) **c)** Build a Tip Calculator app using several JavaFX components and learn how to respond to user interactions with the GUI
9	a) Write a java program that connects to a database using JDBC b) Write a java program to connect to a database using JDBC and insert values into it. c) Write a java program to connect to a database using JDBC and delete values from it
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Reference Books:	
1	The complete Reference Java, 11thedition, Herbert Schildt,TMH
2	Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
23BIT2106	**SEC**	**--**	**1**	**2**	**2**	**30**	**70**	**3 Hrs.**

PYTHON PROGRAMMING

(Common to IT, AIDS and CSBS)

Course Objectives: The major objective of this course is to

1	Introduce core programming concepts of Python programming language
2	Demonstrate about Python data structures like Lists, Tuples, Sets and dictionaries
3	Implement Functions, Modules and Regular Expressions in Python Programming and to create practical and contemporary applications using these

Course Outcomes: After completion of the course student will be able to

S.No	Outcome	Knowledge Level
1	**Demonstrate** Basic Python Programming Concepts for solving problems	K3
2	**Analyze** Python Functions and Modules for real time applications	K4
3	**Evaluate** data collection concepts on Data Handling.	K4
4	**Make use** of NumPy, Pandas for data preprocessing.	K3

SYLLABUS

Unit-I	History of Python Programming Language, Thrust Areas of Python, Installing Anaconda Python Distribution, Installing and Using Jupyter Notebook. Parts of Python Programming Language: Identifiers, Keywords, Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types, Indentation, Comments, Reading Input, Print Output, Type Conversions, the type () Function and Is Operator, Dynamic and Strongly Typed Language. Control Flow Statements: if statement, if-else statement, if...elif…else, Nested if statement, while Loop, for Loop, continue and break Statements, Catching Exceptions Using try and except Statement.
	Sample Experiments:
1	Write a program to find the largest element among three Numbers.
2	Write a Program to display all prime numbers within an interval
3	Write a program to swap two numbers without using a temporary variable.
4	Demonstrate the following Operators in Python with suitable examples. i) Arithmetic Operators ii) Relational Operators iii) Assignment Operators iv) Logical Operators v) Bit wise Operators vi) Ternary Operator vii) Membership Operators viii) Identity Operators
5	Write a program to add and multiply complex numbers
6	Write a program to print multiplication table of a given number

Unit-II	Functions: Built-In Functions, Commonly Used Modules, Function Definition and Calling the function, return Statement and void Function, Scope and Lifetime of Variables, Default Parameters, Keyword Arguments, *args and **kwargs, Command Line Arguments. Strings: Creating and Storing Strings, Basic String Operations, Accessing Characters in String by Index Number, String Slicing and Joining, String Methods, Formatting Strings. Lists: Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-In Functions Used on Lists, List Methods, del Statement
	Sample Experiments:
7	Write a program to define a function with multiple return values.
8	Write a program to define a function using default arguments.
9	Write a program to find the length of the string without using any library functions.
10	Write a program to check if the substring is present in a given string or not.
11	Write a program to perform the given operations on a list: i. Addition ii. Insertion iii. slicing
12	Write a program to perform any 5 built-in functions by taking any list.
Unit-III	Dictionaries: Creating Dictionary, Accessing and Modifying key:value Pairs in Dictionaries, Built-In Functions Used on Dictionaries, Dictionary Methods, del Statement. Tuples and Sets: Creating Tuples, Basic Tuple Operations, tuple() Function, Indexing and Slicing in Tuples, Built-In Functions Used on Tuples, Relation between Tuples and Lists, Relation between Tuples and Dictionaries, Using zip() Function, Sets, Set Methods, Frozenset.
	Sample Experiments:
13	Write a program to create tuples (name, age, address, college) for at least two members and concatenate the tuples and print the concatenated tuples
14	Write a program to count the number of vowels in a string (No control flow allowed).
15	Write a program to check if a given key exists in a dictionary or not
16	Write a program to add a new key-value pair to an existing dictionary
17	Write a program to sum all the items in a given dictionary
Unit-IV	Files: Types of Files, Creating and Reading Text Data, File Methods to Read and Write Data, Reading and Writing Binary Files, Pickle Module, Reading and Writing CSV Files, Python os and os.path Modules. Object-Oriented Programming: Classes and Objects, Creating Classes in Python, Creating Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes Vs Data Attributes, Encapsulation, Inheritance, Polymorphism.
	Sample Experiments:
18	Write a program to sort words in a file and put them in another file. The output file should have only lower-case words, so any upper-case words from source must be lowered
19	Python program to print each line of a file in reverse order.
20	Python program to compute the number of characters, words and lines in a file.
21	Write a program to create, display, append, insert and reverse the order of the items in the array
22	Write a program to add, transpose and multiply two matrices.

23	Write a Python program to create a class that represents a shape. Include methods to calculate its area and perimeter. Implement subclasses for different shapes like circle, triangle, and square.
Unit-V	Introduction to Data Science: Functional Programming, JSON and XML in Python, NumPy with Python, Pandas.
	Sample Experiments:
24	Python program to check whether a JSON string contains complex object or not.
25	Python Program to demonstrate NumPy arrays creation using array () function.
26	Python program to demonstrate use of ndim, shape, size, dtype
27	Python program to demonstrate basic slicing, integer and Boolean indexing
28	Python program to find min, max, sum, cumulative sum of array
29	Create a dictionary with at least five keys and each key represent value as a list where this list contains at least ten values and convert this dictionary as a pandas data frame and explore the data through the data frame as follows: a) Apply head () function to the pandas data frame b) Perform various data selection operations on Data Frame
30	Select any two columns from the above data frame, and observe the change in one attribute with respect to other attribute with scatter and plot operations in matplotlib
Reference Books:	
1	Gowrishankar S, Veena A., Introduction to Python Programming, CRC Press.
2	Python Programming, S Sridhar, J Indumathi, V M Hariharan, 2nd Edition, Pearson, 2024
3	Introduction to Programming Using Python, Y. Daniel Liang, Pearson

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC2103	**MC**	**2**	**--**	**--**	**--**	**30**	**--**	**3 Hrs**

FUNDAMENTALS OF ECONOMICS AND MANAGEMENT

(For CSBS)

Course Objectives: Students are expected to

1.	Understand the basic concepts of Economics at the Micro and Macro level
2.	Familiarize about the Forms of Business Organizations, their Features, Merits & demerits
3.	Understand about basic principles of Management, HR function, Production function and Estimating BEP
4.	Learn about the basics of Finance function and Preparation of Financial Statements
5.	Know the concept of Marketing function and Entrepreneurship

Course Outcomes: At the end of the course the student will be able to

S.No	Outcome	Knowledge Level
1.	**Interpreting** the importance of Economics, demand analysis and Elasticity of demand	K2
2.	**Describe** about the different forms of Business organizations	K2
3.	**Illustrate** the basic concepts of Management, HR function, Production function and estimating the BEP	K2
4.	**Explain** the Finance function and preparation of Financial statements	K2
5.	**Illustrate** the Marketing function and Entrepreneurship	K2

SYLLABUS

UNIT-I (10 Hrs)	**Fundamentals of Economics:** Wealth, Welfare and Scarce Definitions of Economics; Micro and Macro Economics; Demand- Law of Demand, Elasticity of Demand, Types of Elasticity and Factors determining price elasticity of Demand – Law of Diminishing Marginal Utility.
UNIT-II (8 Hrs)	**Forms of Business Organizations**: Features, merits and demerits of Sole Proprietorship, Partnership and Joint Stock Company- Public Enterprises and their types.
UNIT-III (12 Hrs)	**Introduction to Management:** Functions of Management- Taylor's Scientific Management; Henry Fayol's Principles of Management; Human Resource Management – Basic functions of Human Resource Management (in brief). Production Management: Production Planning and Control, Plant Location, Break-Even Analysis- Assumptions, limitations and applications.

UNIT-IV (8 Hrs)	**Financial Management:** Types of Capital: Fixed and Working Capital and Methods of Raising Finance; Final Accounts- Trading Account, Statement of Profit and Loss and Balance Sheet (simple problems).
UNIT-V (8 Hrs)	**Marketing Management and Entrepreneurship:** Marketing Management: Functions of marketing and Distribution Channels. Entrepreneurship: Definition, Characteristics and Functions of an Entrepreneurship.
Text Books:	
1.	S.N. Maheswari, S.K. Maheswari, Financial Accounting Fifth Edition, Vikas Publishing House Pvt. Ltd., New Delhi, 2012
2.	S.C. Sharma and Banga T. R., Industrial Organization & Engineering Economics, Khanna Publications, Delhi-6, 2006
3.	A.R. Arya Sri, Managerial Economics and Financial Analysis, TMH Publications, New Delhi, 2014
4.	Management Science, A.R. Aryasri, Mc Graw Hill Publications, New Delhi
Reference Books:	
1.	G. Sudarshan Reddy – Financial Management - Himalaya Publishing House
2.	P. Subba Rao – Personnel and Human Resource Management –Himalaya Publishing House
3.	Philip Kotler – Marketing Management - Pearson Prentice Hall.

Regulation: R23 | **II / IV - B.Tech. II - Semester**

COMPUTER SCIENCE AND BUSINESS SYSTEMS

COURSE STRUCTURE
(With effect from 2023-24 admitted Batch onwards)

Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23HS2202	Financial Management	HS	2	0	0	2	30	70	100
B23BS2201	Probability & Statistics	ES	3	0	0	3	30	70	100
B23CB2201	Advanced Data Structures & Algorithm Analysis	PC	3	0	0	3	30	70	100
B23IT2202	Software Engineering	PC	3	0	0	3	30	70	100
B23IT2203	Operating Systems	PC	3	0	0	3	30	70	100
B23IT2204	Operating Systems & Software Engineering Lab	PC	0	0	3	1.5	30	70	100
B23CB2202	Advanced Data Structures and Algorithms Lab	PC	0	0	3	1.5	30	70	100
B23CB2203	Full Stack development – 1	SEC	0	1	2	2	30	70	100
B23CB2204	Design Thinking & Innovation	ES	1	0	2	2	30	70	100
B23MC2202	Environmental Science	MC	2	0	0	-	30	-	30
		TOTAL	**17**	**1**	**10**	**21**	**300**	**630**	**930**

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS2202	**HS**	**2**	**--**	**--**	**2**	**30**	**70**	**3 Hrs.**

FINANCIAL MANAGEMENT

(For CSBS)

Course Objectives: Students are expected to

1.	Learn about the importance of Financial Management and Estimating the Present and Future values of money
2.	Familiarize about the valuation of Bond, Preferred stock and Risk & Return
3.	Understand about the concept of leverage in Financial Management, Estimation of cost of capital
4.	Learn about the techniques of Capital Budgeting Decision and Working Capital Management
5.	Know the concept of Cash Management and Receivables Management

Course Outcomes: At the end of the course the student will be able to

S.No	Outcome	Knowledge Level
1.	**Demonstrate** the importance of Financial Management and Estimating the Present and Future values of money	K3
2.	**Describe** about the Valuation of Bonds, Preferred stock and Risk & Return	K2
3.	**Illustrate** the importance of Financial and operating leverages and estimation of cost of capital	K2
4.	**Solve** the Capital Budgeting decisions and Working Capital Management decisions	K3
5.	**Illustrate** the Cash management and Receivables management	K2

SYLLABUS

UNIT-I (10 Hrs)	**Introduction:** Introduction to Financial Management - Goals of the firm - Financial Environments. **Time Value of Money:** Simple and Compound Interest Rates, Amortization, Computing more that once a year, Annuity Factor.
UNIT-II (8 Hrs)	**Valuation of Securities:** Bond Valuation, Preferred Stock Valuation, Common Stock Valuation, Concept of Yield and YTM. **Risk & Return:** Defining Risk and Return, Using Probability Distributions to Measure Risk, Attitudes Toward Risk, Risk and Return in a Portfolio Context, Diversification, The Capital Asset Pricing Model (CAPM).
UNIT-III (12 Hrs)	**Operating & Financial Leverage:** Operating Leverage, Financial Leverage, Total Leverage, Indifference Analysis in leverage study. **Cost of Capital:** Concept, Computation of Specific Cost of Capital for Equity - Preference – Debt, Weighted

	Average Cost of Capital – Factors affecting Cost of Capital.
UNIT-IV (8 Hrs)	**Capital Budgeting:** The Capital Budgeting Concept & Process - An Overview, Generating Investment Project Proposals, Estimating Project, After Tax Incremental Operating Cash Flows, Capital Budgeting Techniques, Project Evaluation and Selection– Alternative Methods **Working Capital Management:** Overview, Working Capital Issues, Financing Current Assets (Short Term and Long Term- Mix), Combining Liability Structures and CurrentAsset Decisions, Estimation of Working Capital.
UNIT-V (8 Hrs)	**Cash Management:** Motives for Holding cash, Speeding Up Cash Receipts, Slowing Down Cash Payouts, Electronic Commerce, Outsourcing, Cash Balances to maintain, Factoring. **Accounts Receivable Management:** Credit & Collection Policies, Analyzing the Credit Applicant, Credit References, Selecting optimum Credit period.

Text Books:	
1.	Prasanna Chandra- Financial Management - Theory & Practice, Tata McGraw Hill.
2.	I M Pandey– Financial Management – Vikas Publishing House Pvt Ltd, New Delhi
3.	G. Sudarshan Reddy– Financial Management - Himalaya Publishing House
Reference Books:	
1.	Srivastava, Misra: Financial Management, OUP
2.	Van Horne and Wachowicz: Fundamentals of Financial Management, Prentice Hall/Pearson Education.
3.	Shashi K. Gupta & R.K. Sharma – Financial Management theory and practice, Kalyani Publishers

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23BS2201	**ES**	**3**	**0**	**0**	**3**	**30**	**70**	**3 Hrs.**

PROBABILITY AND STATISTICS

(Common to AIML, CSE, CSBS, CSIT and IT)

Course Objectives: Students are expected to

1.	Acquire critical thinking skills from the concepts of Descriptive Statistics, Data Science and Probability.
2.	Predict the relationship and the impact of the relationship between the variables.
3.	Obtain Decision-making skills from Sampling theory.

Course Outcomes: Upon the successful completion of this course, Students will

S.No.	Outcome	Knowledge Level
1.	**Compute** various statistical measures like central tendency and spread values.	K2
2.	**Use** the concepts of probability and random variables to solve simple problems based on discrete and continuous probability distributions.	K3
3.	**Determine** correlation and regression coefficients and model a best suitable curve for a given data using the method of least squares.	K3
4.	**Apply** the procedures of sampling theory to find point and interval estimates for various sampling distributions.	K3
5.	**Model** a framework by testing of hypothesis for getting inferences about Population Parameters based on Sample statistic.	K3

SYLLABUS

UNIT-I (10 Hrs.)	**Descriptive statistics and methods for data science:** Data science- Statistics Introduction- Population vs Sample –Collection of data: primary and secondary data- Type of variables: dependent and independent, Categorical and Continuous variables- Data visualization- Measures of Central tendency- Measures of Variability (spread or variance)- Moments- Measures of Skewness and Kurtosis.
UNIT-II (12 Hrs.)	**Probability, Random variables and Distributions:** Probability– Conditional probability, addition, Multiplication theorems and Baye's theorem. Random variables – Discrete and Continuous random variables – Distribution functions – Probability mass function, Probability density function and Cumulative distribution functions – Mathematical Expectation – Variance. **Discrete and Continuous Distributions:** **Discrete Distributions**: Bernoulli, Binomial and Poisson distributions - Mean, Variance, Fitting of distributions.

	Continuous Distributions: Uniform distribution, Normal Distribution, Standard Normal Variate - Mean, Variance.
UNIT-III (12 Hrs.)	**Correlation, Regression andCurve fitting**: **Correlation**: Definition, Karl Pearson's Coefficient of Correlation, Limits for correlation coefficient, Rank Correlation, Spearman's rank correlation coefficient (without proofs). **Linear Regression**: Regression- Regression coefficients and properties–Multiple Linear Regression. **Curve Fitting**: Method of Least Squares, Fitting of a Straight line, Fitting of a Parabola, Fitting of an Exponential curves: $y = ae^{bx}, y = ab^x$ and Power curve: $y = ax^b$.
UNIT-IV (8 Hrs.)	**Sampling Theory:** Introduction – Population and Samples –Parameter and Statistic- Sampling distribution of statistic-Standard error- Sampling distribution of Means and Variance (definition only) – Point and Interval estimations. Maximum error of estimate – Central limit theorem (without proof) – Estimation using t, χ^2 and F-distributions.
UNIT-V (12 Hrs.)	**Testing of Hypothesis:** Testing of Hypothesis- Formulation of Null hypothesis, Alternative hypothesis, Critical region, level of significance, Errors in sampling- Type-I-error, Type-II-error, One-tailed and Two-tailed tests, Degrees of freedom. **Large Sample Theory:** Test of significance for single and difference of Proportions. **Small Sample Theory:** Student's-t-distribution: Definition, t-test for single mean, t-test for difference of means, Paired t-test for difference of means. F-distribution: Definition, F-test for equality of two population variances. Chi-square distribution: Definition, Chi-square test for attributes and goodness of fit.

Text Books:	
1.	Fundamentals of Mathematical Statistics by S. C. Gupta and V.K. Kapoor, Sultan Chand & Sons Publishers.
2	Probability and Statistics for Engineers, Miller and Freund, 7th edition, Prentice-Hall India.
Reference Books:	
1.	Probability and statistics for Engineers and Scientists by Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye, Eighth edition, Pearson Education.
2.	Probability and Statistics for Engineering and the Sciences, Jay l. Devore, 8th Edition, Cengage.
3.	Introduction to probability and statistics Engineers and the Scientists, Sheldon M. Ross, 4th Edition, Academic Foundation, 2011.
4.	Johannes Ledolter and Robert V. Hogg, Applied statistics for Engineers and Physical

	Scientists, 3rd Edition, Pearson, 2010.
5.	Probability, Statistics and Random Processes by T.Veerarajan, Tata McGraw Hill Pub.
6.	Higher Engineering Mathematics, by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers
7.	Paul L. Meyer, Introductory Probability and Statistical Applications (2nd edn.), Addison-Wesley, 1970.
e-Resources:	
1.	http://www.swayam.gov.in
2.	https://archive.nptel.ac.in/courses/106/104/106104233/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CB2201	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

ADVANCED DATA STRUCTURES & ALGORITHM ANALYSIS

(For CSBS)

Course Objectives: The major objective of this course is to

1.	Provide knowledge on advance data structures frequently used in Computer Science domain
2.	Develop skills in algorithm design techniques popularly used
3.	Understand the use of various data structures in the algorithm design

Course Outcomes: After completion of the course student will be able to

S.No	Outcome	Knowledge Level
1.	**Apply** non-linear data structures for solving a given problem.	K3
2.	**Apply** divide and conquer technique to solve problems and calculate time complexity.	K3
3.	**Apply** different design technique algorithms like greedy, dynamic programming, backtracking and branch and bound for solving problems.	K3
4.	**Apply** the knowledge of complexity classes P, NP and NP-Complete for solving graph and Scheduling problems	K3

SYLLABUS

UNIT-I (10Hrs)	AVL Trees – Creation, Insertion, Deletion operations and Applications B-Trees – Creation, Insertion, Deletion operations and Applications Heap Trees (Priority Queues) – Min and Max Heaps, Operations and Applications
UNIT-II (10 Hrs)	Graphs – Terminology, Representations, Basic Search and Traversals, Connected Components and Biconnected Components, applications Introduction to Algorithm Analysis, Space and Time Complexity analysis, Asymptotic Notations. Divide and Conquer: The General Method, Quick Sort, Merge Sort, Strassen's matrix multiplication, Convex Hull
UNIT-III (10 Hrs)	Greedy Method: General Method, Job Sequencing with deadlines, Knapsack Problem, Minimum cost spanning trees, Single Source Shortest Paths Dynamic Programming: General Method, All pairs shortest paths, Single Source Shortest Paths– General Weights (Bellman Ford Algorithm), Optimal Binary Search Trees, 0/1 Knapsack, String Editing, Travelling Salesperson problem

UNIT-IV (10 Hrs)	Backtracking: General Method, 8-Queens Problem, Sum of Subsets problem, Graph Coloring, 0/1 Knapsack Problem Branch and Bound: The General Method, 0/1 Knapsack Problem, Travelling Salesperson problem
UNIT-V (10 Hrs)	NP Hard and NP Complete Problems: Basic Concepts, Cook's theorem NP Hard Graph Problems: Clique Decision Problem (CDP), Chromatic Number Decision Problem (CNDP), Traveling Salesperson Decision Problem (TSP) NP Hard Scheduling Problems: Scheduling Identical Processors, Job Shop Scheduling
Textbooks:	
1.	Fundamentals of Data Structures in C++, Horowitz, Ellis; Sahni, Sartaj; Mehta, Dinesh, 2nd Edition Universities Press
2.	Computer Algorithms in C++, Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, 2nd Edition University Press
Reference Books:	
1.	Data Structures and program design in C, Robert Kruse, Pearson Education Asia
2.	An introduction to Data Structures with applications, Trembley & Sorenson, McGraw Hill
3.	The Art of Computer Programming, Vol.1: Fundamental Algorithms, Donald E Knuth, Addison-Wesley, 1997.
4.	Data Structures using C & C++: Langsam, Augenstein & Tanenbaum, Pearson, 1995
5.	Algorithms + Data Structures & Programs:, N.Wirth, PHI
6.	Fundamentals of Data Structures in C++: Horowitz Sahni & Mehta, Galgottia Pub.
7.	Data structures in Java:, Thomas Standish, Pearson Education Asia
e-Resources	
1.	https://www.tutorialspoint.com/advanced_data_structures/index.asp
2.	http://peterindia.net/Algorithms.html
3.	Abdul Bari, Introduction to Algorithms (youtube.com)

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2202	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

SOFTWARE ENGINEERING

(Common to IT and CSBS)

Course Objectives: The major objective of this course is to

1.	Exposure to different phases of software development, including Waterfall, Unified Process, and agile process models.
2.	Hands-on experience with various software engineering practices, such as requirements analysis, Course Code analysis, debugging, testing, traceability, and version control.
3.	Exposure to software design techniques.

Course Outcomes: At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	**Apply** software engineering concepts to illustrate various phases of a lifecycle and choose appropriate process model depending on the user requirements.	K3
2.	**Apply** software project management concepts to prepare software requirements specification report.	K3
3.	**Apply** various software architectures and design-patterns to select the appropriate techniques to design the software and User Interface.	K3
4.	**Examine** whether all the requirements specified have been achieved or not under specified quality standards.	K3
5.	**Apply** Computer-Aided Software Engineering (CASE) tools for managing projects.	K3

SYLLABUS

UNIT-I (10Hrs)	**Introduction:** Evolution, Software development projects, Exploratory style of software developments, Emergence of software engineering, Notable changes in software development practices, Computer system engineering. **Software Life Cycle Models:** Basic concepts, Waterfall model and its extensions, Rapid application development, Agile development model, Spiral model.
UNIT-II (10 Hrs)	**Software Project Management:** Software project management complexities, Responsibilities of a software project manager, Metrics for project size estimation, Project estimation techniques, Empirical Estimation techniques, COCOMO, Halstead's software science, risk management. **Requirements Analysis And Specification:** Requirements gathering and analysis, Software Requirements Specification (SRS), Formal system specification, Axiomatic specification, Algebraic specification, Executable specification and 4GL.

UNIT-III (10 Hrs)	**Software Design:** Overview of the design process, how to characterize a good software design? Layered arrangement of modules, Cohesion and Coupling approaches to software design. **Agility:** Agility and the Cost of Change, Agile Process, Extreme Programming (XP), Other Agile Process Models. (Text Book 2) **Function-Oriented Software Design:** Overview of SA/SD methodology, Structured analysis, Developing the DFD model of a system, Structured design, Detailed design, and Design Review. **User Interface Design:** Characteristics of a good user interface, Basic concepts, Types of user interfaces, Fundamentals of component-based GUI development, and user interface design methodology.
UNIT-IV (10 Hrs)	**Coding and Testing:** Coding, Course Code review, Software documentation, Testing, Black-box testing, White-Box testing, Debugging, Program analysis tools, Integration testing, testing object-oriented programs, Smoke testing, and Some general issues associated with testing. **Software Reliability and Quality Management:** Software reliability. Statistical testing, Software quality, Software quality management system, ISO 9000.SEI Capability maturity model. Few other important quality standards, and Six Sigma.
UNIT-V (10 Hrs)	**Computer-Aided Software Engineering (Case):** CASE and its scope, CASE environment, CASE support in the software life cycle, other characteristics of CASE tools, Towards second generation CASE Tool, and Architecture of a CASE Environment. **Software Maintenance:** Characteristics of software maintenance, Software reverse engineering, Software maintenance process models and Estimation of maintenance cost. **Software Reuse:** reuse- definition, introduction, reason behind no reuse so far, Basic issues in any reuse program, A reuse approach, and Reuse at organization level.

Textbooks:	
1.	Fundamentals of Software Engineering, Rajib Mall, 5th Edition, PHI.
2.	Software Engineering A practitioner's Approach, Roger S. Pressman, 9th Edition, Mc-Graw Hill International Edition.
Reference Books:	
1.	Software Engineering, Ian Sommerville,10thEdition, Pearson.
2.	Software Engineering, Principles and Practices, Deepak Jain, Oxford University Press.
e-Resources	
1.	https://nptel.ac.in/courses/106/105/106105182/
2.	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01260589506387148827_shared/overview
3.	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013382690411003904735_shared/overview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2203	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

OPERATING SYSTEMS

(Common to IT and CSBS)

Course Objectives: The major objective of this course is to

1.	Understand the basic concepts and principles of operating systems, including process management, memory management, file systems, and Protection.
2.	Make use of process scheduling algorithms and synchronization techniques to achieve better performance of a computer system.
3.	Illustrate different conditions for deadlock and their possible solutions.

Course Outcomes: After completion of the course student will be able to

S.No	Outcome	Knowledge Level
1.	**Describe** Operating System features and system structures.	K2
2.	**Apply** CPU Scheduling Algorithms for Multi Process and Multi Threaded Operating systems.	K3
3.	**Solve** Process Synchronization problems to avoid occurrence of Deadlock situations	K3
4.	**Apply** various Memory Management Schemes for Primary and Secondary memory.	K3
5.	**Describe** file Operations and protection methods.	K2

SYLLABUS

UNIT-I (10Hrs)	**Operating Systems Overview:** Introduction, Operating system functions, Operating systems operations, Computing environments, Free and Open-Source Operating Systems **System Structures:** Operating System Services, User and Operating-System Interface, system calls, Types of System Calls, system programs, Operating system Design and Implementation, Operating system structure, Building and Booting an Operating System, Operating system debugging.
UNIT-II (10 Hrs)	**Processes:** Process Concept, Process scheduling, Operations on processes, Inter-process communication. **Threads and Concurrency:** Multithreading models, Thread libraries, Threading issues. **CPU Scheduling:** Basic concepts, Scheduling criteria, Scheduling algorithms, Multiple processor scheduling.
UNIT-III (10 Hrs)	**Synchronization Tools:** The Critical Section Problem, Peterson's Solution, Mutex Locks, Semaphores, Monitors, Classic problems of Synchronization.

	Deadlocks: system Model, Deadlock characterization, Methods for handling Deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from Deadlock.
UNIT-IV (10 Hrs)	**Memory-Management Strategies:** Introduction, Contiguous memory allocation, Paging, Structure of the Page Table, Swapping. **Virtual Memory Management:** Introduction, Demand paging, Copy-on-write, Page replacement, Allocation of frames, Thrashing. **Storage Management:** Overview of Mass Storage Structure, HDD Scheduling.
UNIT-V (10 Hrs)	**File System:** File System Interface: File concept, Access methods, Directory Structure. **File system Implementation:** File-system structure, File-system Operations, Directory implementation, Allocation method, Free space management. **File-System Internals:** FileSystem Mounting, Partitions and Mounting, File Sharing. **Protection:** Goals of protection, Principles of protection, Protection Rings, Domain of protection, Access matrix.
Textbooks:	
1.	Operating System Concepts, Silberschatz A, Galvin P B, Gagne G, 10th Edition Wiley, 2018.
2.	Modern Operating Systems, Tanenbaum A S, 4th Edition, Pearson , 2016.
Reference Books:	
1.	Operating Systems -Internals and Design Principles, Stallings W, 9th edition, Pearson, 2018.
2.	Operating Systems: A Concept Based Approach, D.M Dhamdhere, 3rd Edition, McGraw- Hill, 2013.
3.	Nutt G, Operating Systems, 3rd edition, Pearson Education, 2004.
e-Resources	
1.	https://nptel.ac.in/courses/106106144
2.	https://peterindia.net/OperatingSystems.html

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2204	PC	--	--	3	1.5	30	70	3 Hrs.

OPERATING SYSTEMS & SOFTWARE ENGINEERING LAB

(Common to IT and CSBS)

Course Objectives: The major objective of this course is to

1	Provide insights into system calls, file systems, semaphores.
2	Develop and debug CPU Scheduling algorithms, page replacement algorithms, thread implementation
3	Implement Bankers Algorithms to Avoid the Dead Lock
4	acquire the generic software development skill through various stages of software life cycle
5	generate test cases for software testing

Course Outcomes: At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1	**Make use of** Unix commands and System calls to explore the operating system functionalities**.**	K3
2	**Apply** various Operating System services like process scheduling & synchronization, memory management, deadlock handling and File management.	K3
3	**Apply** Object oriented concepts for Software Development Life Cycle using Rational Rose.	K3
4	**Design** UML diagrams for any real world application.	K4

SYLLABUS

Experiments covering the Topics:

- UNIX fundamentals, commands & system calls
- CPU Scheduling algorithms, thread processing
- IPC, semaphores, monitors, deadlocks
- Page replacement algorithms, file allocation strategies
- Memory allocation strategies
- Software Requirement Specification, DFD, CFD
- Software estimation, UML diagrams, test case design

Sample Experiments in Operating Systems:

1	Practicing of Basic UNIX Commands (ex: cat, touch, cp, ls, mv, rm, mkdir, rmdir, grep, sort, cut, wc, chmod, sed, awk)
2	Write programs using the following UNIX operating system calls fork, vfork, exec, getpid, wait, close, read and write

3	Simulate UNIX commands like cp, ls and grep
4	Simulate the following CPU scheduling algorithms a) FCFS b) SJF c) Round Robin
5	Write a program to illustrate concurrent execution of threads using pthreads library.
6	Write a program to solve producer-consumer problem using Semaphores.
7	Implement Bankers Algorithm for Dead Lock avoidance
8	Implement the following memory allocation methods a) MFT b) MVT
9	Simulate the following page replacement algorithms a) FIFO b) LRU
10	Simulate the following file allocation strategies a) Sequential b) Indexed c) Linked
11	Download and install nachos operating system and experiment with it
Sample Experiments in Software Engineering:	
1	Perform the following, for the following experiments: I. Do the Requirement Analysis and Prepare SRS II. Draw E-R diagrams, DFD, CFD and structured charts for the project. a. Course Registration System b. Students Marks Analyzing System c. Online Ticket Reservation System d. Stock Maintenance
2	Consider any application, using COCOMO model, estimate the effort.
3	Consider any application, Calculate effort using FP oriented estimation model.
4	Draw the UML Diagrams for the problem a, b, c, d.
5	Design the test cases for e-Commerce application (Flipkart, Amazon)
6	Design the test cases for a Mobile Application (Consider any example from App store)
7	Design and Implement ATM system through UML Diagrams.
Reference Books:	
1	“Unix Shell Programming” by Yashwant P kanitkar, BPB Publications.
2	“Introduction to UNIX & SHELL programming”, by M.G. Venkatesh Murthy, Pearson Education.
3	“Operating System Concepts”, Silberschatz A, Galvin P B, and Gagne G, 9th edition, Wiley, 2013.
4	“Let Us C: Authentic guide to C programming language“, by Yashavant Kanetkar, 19th Edition
5	“Unified Modeling Language User Guide” by Booch, Pearson Education.
6	Object Oriented Software Engineering using UML, Patrons and Java by Bernd Bruegge, Allen H. Dutoit, 3rd Edition, Prentice Hall Press

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CB2202	**PC**	**--**	**--**	**3**	**1.5**	**30**	**70**	**3 Hrs.**

ADVANCED DATA STRUCTURES & ALGORITHMS LAB

(For CSBS)

Course Objectives: The major objective of this course is to

1	Acquire practical skills in constructing and managing Data structures
2	Apply the popular algorithm design methods in problem-solving scenarios

Course Outcomes: After completion of the course student will be able to

S. No	Outcome	Knowledge Level
1	**Apply** Non-Linear data structures for solving any given problem.	K3
2	**Apply** divide and conquer technique for solving quick and merge sort problems.	K3
3	**Apply** greedy method for solving Job sequencing and Single source shortest path problems.	K3
4	**Use** dynamic programming, Backtracking and branch and bound for solving given problems.	K3

SYLLABUS

1	Construct an AVL tree for a given set of elements which are stored in a file. And implement insert and delete operation on the constructed tree. Write contents of tree into a new file using in-order.
2	Construct B-Tree an order of 5 with a set of 100 random elements stored in array. Implement searching, insertion and deletion operations.
3	Construct Min and Max Heap using arrays, delete any element and display the content of the Heap
4	Implement BFT and DFT for given graph, when graph is represented by a) Adjacency Matrix b) Adjacency Lists
5	Write a program for finding the biconnected components in a given graph.
6	Implement Quick sort and Merge sort and observe the execution time for various input sizes (Average, Worst and Best cases).
7	Compare the performance of Single Source Shortest Paths using Greedy method when the graph is represented by adjacency matrix and adjacency lists.
8	Implement Job Sequencing with deadlines using Greedy strategy.
9	Write a program to solve 0/1 Knapsack problem Using Dynamic Programming.
10	Implement N-Queens Problem Using Backtracking.
11	Use Backtracking strategy to solve 0/1 Knapsack problem.
12	Implement Travelling Salesperson problem using Branch and Bound approach.

Reference Books:	
1	Fundamentals of Data Structures in C++, Horowitz Ellis, Sahni Sartaj, Mehta, Dinesh, 2nd Edition, Universities Press
2	Computer Algorithms/C++ Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, 2nd Edition, University Press
3	Data Structures and program design in C, Robert Kruse, Pearson Education Asia
4	An introduction to Data Structures with applications, Trembley & Sorenson, McGraw Hill

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CB2203	SEC	--	1	2	2	30	70	3 Hrs.

FULL STACK DEVELOPMENT-1

(For CSBS)

Course Objectives: The main objectives of the course are to

1	Make use of HTML elements and their attributes for designing static web pages.
	Build a web page by applying appropriate CSS styles to HTML elements
2	Experiment with JavaScript to develop dynamic web pages and validate forms

Course Outcomes: After completion of the course, the student will be able to

S. No	Outcome	Knowledge Level
1	Develop static web pages using HTML and Cascading Style Sheet	K3
2	Develop a dynamic web page for validations using JavaScript	K3
3	Model a Basic web server using node.js	K3

SYLLABUS

1	**Lists, Links and Images** a. Write a HTML program, to explain the working of lists. Note: It should have an ordered list, unordered list, nested lists and ordered list in an unordered list and definition lists. b. Write a HTML program, to explain the working of hyperlinks using <a> tag and href, target Attributes. c. Create a HTML document that has your image and your friend's image with a specific height and width. Also when clicked on the images it should navigate to their respective profiles. d. Write a HTML program, in such a way that, rather than placing large images on a page, the preferred technique is to use thumbnails by setting the height and width parameters to something like to 100*100 pixels. Each thumbnail image is also a link to a full sized version of the image. Create an image gallery using this technique
2	**HTML Tables, Forms and Frames** a. Write a HTML program, to explain the working of tables. (use tags: <table>, <tr>, <th>, <td> and attributes: border, rowspan, colspan) b. Write a HTML program, to explain the working of tables by preparing a timetable. (Note: Use <caption> tag to set the caption to the table & also use cell spacing, cell padding, border, rowspan, colspan etc.). c. Write a HTML program, to explain the working of forms by designing Registration form. (Note: Include text field, password field, number field, date of birth field, checkboxes, radio buttons, list boxes using <select>&<option> tags, <text area>

	and two buttons ie: submit and reset. Use tables to provide a better view). d. Write a HTML program, to explain the working of frames, such that page is to be divided into 3 parts on either direction. (Note: first frame image, second frame paragraph, third frame □ hyperlink. And also make sure of using “no frame” attribute such that frames to be fixed).
3	**HTML 5 and Cascading Style Sheets, Types of CSS** a. Write a HTML program, that makes use of <article>, <aside>, <figure>, <figcaption>, <footer>, <header>, <main>, <nav>, <section>, <div>, <span> tags. b. Write a HTML program, to embed audio and video into HTML web page. c. Write a program to apply different types (or levels of styles or style specification formats) - inline, internal, external styles to HTML elements. (identify selector, property and value).
4	**Selector forms** • Write a program to apply different types of selector forms ● Simple selector (element, id, class, group, universal) ● Combinator selector (descendant, child, adjacent sibling, general sibling) ● Pseudo-class selector ● Pseudo-element selector ● Attribute selector
5	**CSS with Color, Background, Font, Text and CSS Box Model** a. Write a program to demonstrate the various ways you can reference a color in CSS. b. Write a CSS rule that places a background image halfway down the page, tilting it horizontally. The image should remain in place when the user scrolls up or down. c. Write a program using the following terms related to CSS font and text: i. font-size ii. font-weight iii. font-style iv. text-decoration v. text-transformation vi. text-alignment d. Write a program, to explain the importance of CSS Box model using i. Content ii. Border iii. Margin iv. padding
6	**Applying JavaScript - internal and external, I/O, Type Conversion** a. Write a program to embed internal and external JavaScript in a web page. b. Write a program to explain the different ways for displaying output. c. Write a program to explain the different ways for taking input. d. Create a webpage which uses prompt dialogue box to ask a voter for his name and age. Display the information in table format along with either the voter can vote or not
7	**JavaScript Pre-defined and User-defined Objects** a. Write a program using document object properties and methods. b. Write a program using window object properties and methods. c. Write a program using array object properties and methods. d. Write a program using math object properties and methods. e. Write a program using string object properties and methods. f. Write a program using regex object properties and methods.

	g. Write a program using date object properties and methods. h. Write a program to explain user-defined object by using properties, methods, accessors , constructors and display.
8	**JavaScript Conditional Statements and Loops** a. Write a program which asks the user to enter three integers, obtains the numbers from the user and outputs HTML text that displays the larger number followed by the words “LARGER NUMBER” in an information message dialog. If the numbers are equal, output HTML text as “EQUAL NUMBERS”. b. Write a program to display week days using switch case . c. Write a program to print 1 to 10 numbers using for, while and do-while loops . d. Write a program to display the denomination of the amount deposited in the bank in terms of 100’s, 50’s, 20’s, 10’s, 5’s, 2’s & 1’s. (Eg: If deposited amount is Rs.163, the output should be 1-100’s, 1-50’s, 1- 10’s, 1-2’s & 1-1’s)
9	**JavaScript Functions and Events** a. Design a appropriate function should be called to display ● Factorial of that number ● Fibonacci series up to that number ● Prime numbers up to that number ● Is it palindrome or not b. Design a HTML having a text box and four buttons named Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate function should be called to display • Factorial of that number • Fibonacci series up to that number • Prime numbers up to that number • Is it palindrome or not
10	Write a program to validate the following fields in a registration page i. Name (start with alphabet and followed by alphanumeric and the length should not be less than 6 characters) ii. Mobile (only numbers and length 10 digits) iii. E-mail (should contain format like xxxxxxx@xxxxxx.xxx)
11	**Node.js** a. Write a program to show the work flow of Java Script Course Code executable by creating web server in Node.js. b. Write a program to transfer data over http protocol using http module c. Createatextfilesrc.txtandaddthefollowingcontenttoit.(HTML,CSS,JavaScript,Node.js) d. Write a program to parse an URL using URL module. e. Writeaprogramtocreateanuser-definedmoduleandshowtheworkflowofModularizationofapplicationusingNode.js
Reference Books:	
1	Programming the World Wide Web, 7th Edition, Robet W Sebesta, Pearson, 2013.

2	Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O'Reilly.
Web Links:	
1.	https://www.w3schools.com/html
2.	https://www.w3schools.com/css
3.	https://www.w3schools.com/js/
4.	https://www.w3schools.com/nodejs
5.	https://www.w3schools.com/typescript

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CB2204	ES	1	--	2	2	30	70	3 Hrs.

DESIGN THINKING & INNOVATION

(Common to all Programmes of Engineering)

Course Objectives:

1.	Bring awareness on innovative design and new product development.
2.	Explain the basics of design thinking.
3.	Familiarize the role of reverse engineering in product development.
4.	Train how to identify the needs of society and convert into demand.
5.	Introduce product planning and product development process.

Course Outcomes: After completion of this course, students will be able to

S.No	Outcome	Knowledge Level
1.	**Define** the concepts related to design thinking.	K1
2.	**Explain** the fundamentals of Design Thinking and innovation.	K2
3.	**Apply** the design thinking techniques for solving problems in various sectors.	K3
4.	**Analyse** to work in a multidisciplinary environment.	K4
5.	**Evaluate** the value of creativity.	K5

SYLLABUS

UNIT-I (10Hrs)	Introduction to elements and principles of Design, basics of design-dot, line, shape, form as fundamental design components. Principles of design. Introduction to design thinking, history of Design Thinking, New materials in Industry.
UNIT-II (10 Hrs)	Design thinking process (empathize, analyze, idea & prototype), implementing the process in driving inventions, design thinking in social innovations. Tools of design thinking - person, costumer, journey map, brainstorming, product development. **Activity:** Every student presents their idea in three minutes, Every student can present design process in the form of flow diagram or flow chart etc. Every student should explain about product development.
UNIT-III (10 Hrs)	Art of innovation, Difference between innovation and creativity, role of creativity and innovation in organizations. Creativity to Innovation. Teams for innovation, Measuring the impact and value of creativity. **Activity:** Debate on innovation and creativity, Flow and planning from idea to innovation, Debate on value-based innovation.
UNIT-IV (10 Hrs)	Problem formation, introduction to product design, Product strategies, Product value, Product planning, product specifications. Innovation towards product design Case studies.

	Activity: Importance of modeling, how to set specifications, Explaining their own product design.
UNIT-V (10 Hrs)	Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business – Business challenges: Growth, Predictability, Change, Maintaining Relevance, Extreme competition, Standardization. Design thinking to meet corporate needs. Design thinking for Startups. Defining and testing Business Models and Business Cases. Developing & testing prototypes. **Activity:** How to market our own product, about maintenance, Reliability and plan for startup.
Textbooks:	
1.	Tim Brown, Change by design, 1/e, Harper Bollins, 2009.
2.	Idris Mootee, Design Thinking for Strategic Innovation, 1/e, Adams Media, 2014.
Reference Books:	
1.	David Lee, Design Thinking in the Classroom, Ulysses press, 2018.
2.	Shrrutin N Shetty, Design the Future, 1/e, Norton Press, 2018.
3.	William lidwell, Kritinaholden, &Jill butter, Universal principles of design, 2/e, Rockport Publishers, 2010.
4.	Chesbrough.H, The era of open innovation, 2003.
e-Resources:	
1.	https://nptel.ac.in/courses/110/106/110106124/
2.	https://nptel.ac.in/courses/109/104/109104109/
3.	https://swayam.gov.in/nd1_noc19_mg60/preview
4.	https://onlinecourses.nptel.ac.in/noc22_de16/preview

Course Code	Category	L	T	P	C	C.I.E	S.E.E	Exam
B23MC2202	**MC**	**2**	**--**	**--**	**--**	**30**	**--**	**--**

ENVIRONMENTAL SCIENCE

(Common to AIDS, AIML, CE, CSBS, CSIT, IT and ME.)

Course Objectives: The objective of the course is to impart:

1.	Overall view on natural resources.
2.	Awareness on ecosystem and its services.
3.	Various environmental challenges induced due to unplanned anthropogenic activities.
4.	Consciousness on the social issues, environmental legislation and global treaties

Course Outcomes: At the end of the course, the students will be able to

S. No	Outcome	Knowledge Level
1	**Describe** natural resources and their interaction	K2
2	**Illustrate** ecosystem types, biodiversity and conservation strategies	K2
3	**Summarize** contaminants of environment and preventive methods	K2
4	**Explain** protection of environment by employing constitutional provisions	K2
5	**Explain** global scenario of surroundings and social conditions	K2

SYLLABUS

UNIT-I (8 Hrs)	**Multidisciplinary Nature of Environmental Studies:** Definition, Scope and Importance – Need for Public Awareness. Natural Resources: Renewable and non-renewable resources – Natural resources and associated problems. Forest resources – Use and over – exploitation, deforestation, case studies – Timber extraction – Mining, dams and other effects on forest and tribal people. Water resources – Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems. Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies. Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. Energy resources: Growing energy needs, renewable and non-renewable energy sources use of alternate energy sources
UNIT-II (8 Hrs)	**Ecosystems:** Concept of an ecosystem. – Structure and function of an ecosystem – Producers, consumers and decomposers – Energy flow in the ecosystem – Ecological succession – Food chains, food webs and ecological pyramids – Introduction, types, characteristic features, structure and function of the following ecosystem: a. Forest ecosystem. b. Grassland ecosystem c. Desert ecosystem d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)

	Biodiversity and Its Conservation : Introduction and Definition - genetic, species and ecosystem diversity – Bio-geographical classification of India – Value of biodiversity: consumptive use, Productive use, social, ethical, aesthetic and option values – Biodiversity at global, National and local levels – India as a mega-diversity nation – Hot-sports of biodiversity – Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – Endangered and endemic species of India – Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.
UNIT-III (6 Hrs)	**Environmental Pollution:** Definition, Cause, effects and control measures of: a. Air Pollution. b. Water pollution c. Soil pollution d. Marine pollution e. Noise pollution f. Thermal pollution g. Nuclear hazards. Solid Waste Management: Causes, effects and control measures of urban and industrial wastes – Role of an individual in prevention of pollution – Pollution case studies – Disaster management: floods, earthquake, cyclone and landslides
UNIT-IV (6 Hrs)	**Social Issues and the Environment:** From Unsustainable to Sustainable development – Urban problems related to energy watershed management – Resettlement and rehabilitation of people; its problems and concerns – Carbon credits, Mission LiFE. Environmental ethics: Issues and possible solutions. Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust –Wasteland reclamation – Consumerism and waste products. Environment Protection Act. – Air (Prevention and Control of Pollution) Act. – Water (Prevention and control of Pollution) Act – Wildlife Protection Act – Forest Conservation Act. Issues involved in enforcement of environmental legislation – Public awareness.
UNIT-V (6 Hrs)	**Human Population And The Environment:** Population growth, variation among nations. Population explosion – Family Welfare Programmes. – Environment and human health – Human Rights – Value Education – HIV/AIDS – Women and Child Welfare – Role of information Technology in Environment and human health – Case studies. Field Work: Visit to a local area to document environmental assets River/forest grassland/hill/mountain – Visit to a local polluted site-Urban/Rural/Industrial/Agricultural Study of common plants, insects, and birds – river, hill slopes, etc.

Text Books:	
1.	Erach Bharucha, Text book of Environmental Studies for Undergraduate Courses, Universities Press (India) Private Limited, 2019.
2.	Palaniswamy, Environmental Studies, 2/e, Pearson education, 2014.
3.	S. Azeem Unnisa, Environmental Studies, Academic Publishing Company, 2021.
4.	K. Raghavan Nambiar, “Text book of Environmental Studies for Undergraduate Courses as per UGC model syllabus”, SciTech Publications (India), Pvt. Ltd, 2010.
5	K. V. S. G. Murali Krishna, The Book of Environmental Studies, Savera Publishing House.

6	Environmental Studies, R. Rajagopalan, 2nd Edition, 2011, Oxford University Press.
Reference Books:	
1.	Deeksha Dave and S.S. Katewa, Textbook of Environmental Studies, 2/e, Cengage Learning.
2.	M. Anji Reddy, "Textbook of Environmental Sciences and Technology", BS Publication, 2014.
3.	J.P. Sharma, Comprehensive Environmental studies, Laxmi publications, 2006.
4.	J. Glynn Henry and Gary W. Heinke, Environmental Sciences and Engineering, Prentice Hall of India Private limited, 1988.
5.	G.R. Chatwal, A Text Book of Environmental Studies, Himalaya Publishing House, 2018
6.	Gilbert M. Masters and Wendell P. Ela, Introduction to Environmental Engineering and Science, 1/e, Prentice Hall of India Private limited, 1991.
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc23_hs155/preview
2.	https://www.edx.org/learn/environmental-science/rice-university-ap-r-environmental-science-part-3-pollution-and-resources?index=product&objectID=course-3a6da9f2-lec07.pdf (iasri.res.in)
3.	http://ecoursesonline.iasri.res.in/Courses/Environmental%20Science-I/Data%20Files/pdf/lec07.pdf
4.	https://www.youtube.com/watch?v=5QxxaVfgQ3k